

MATH 2072U

Final solutions, 2024

Instructor: L. van Veen

Apr. 14, 2024

1. [15 marks] Consider the equation

$$\ln(x) - \frac{1}{2}x + 1 = 0$$

- (a) Show that this equation has a unique solution in the domain $[2, 6]$.

First, we show that there is at least one solution. This function is continuously differentiable on the domain and $f(2)f(6) < 0$, so there is at least one solution. Next we establish that the function is monotonically decreasing and therefore the solution is unique (as a consequence of the intermediate value theorem). The derivative is $1/x - 1/2$. This function is monotonically decreasing for $x > 0$, and at $x = 2$ it equals 0. Consequently, the derivative is non-positive on the whole domain, the function is monotonic and the solution is unique.

- (b) Starting from the interval $[a^{(0)}, b^{(0)}] = [2, 6]$, execute two steps of the bisection method.

k	a_k	b_k	m_k	e_k
0	2.0	6.0	4.0	2.0
1	4.0	6.0	5.0	1.0
2	5.0	6.0	5.5	0.5

Here I have taken m_k to be the approximate solution after k steps, with maximal error e_k .

- (c) How many more steps of the bisection algorithm would you need to find an approximate solution with an error less than 10^{-10} ?
 Since the error decreases by a factor of $1/2$ in every iteration, we have $0.5(0.5)^k < 10^{-10}$, from which we find that $k \geq 33$.
- (d) Use Newton iteration, starting from $x_0 = 6$, to find an approximate solution with a residual less than 0.001. At every iteration, list the current approximate solution, the residual and an estimate of the error.

k	x_k	$ f(x_k) $	$ \delta x $
0	6.0	0.2082405	0.6247216
1	5.375278	5.828834×10^{-3}	1.856535×10^{-2}
2	5.356713	5.978272×10^{-6}	1.908050×10^{-5}

Each update step is computed as

$$\delta x = -\frac{2x \ln(x) - x^2 + 2x}{2 - x}$$

- (e) How many more steps of the Newton algorithm would you need to find an approximate solution with a residual less than 10^{-10} ?
 The residuals will decrease approximately quadratically, so we expect that after one more iteration it has order of magnitude 10^{-12} . Therefore, one more iteration should be enough.

2. [10 marks] Consider the following linear system:

$$A\mathbf{x} = \mathbf{b}, \text{ where } A = \begin{bmatrix} 1.0 & -2.5 & -2.0 \\ -4.0 & -1.0 & -0.1 \\ 3.0 & 2.0 & 1.0 \end{bmatrix} \text{ and } \mathbf{b} = \begin{bmatrix} -4.3505 \\ 2.2552 \\ 8.0052 \end{bmatrix}$$

(a) Compute matrices P , L and U such that $PA = LU$.

$$\begin{array}{ccc} \begin{bmatrix} 1.0 & -2.5 & -2.0 \\ -4.0 & -1.0 & -0.1 \\ 3.0 & 2.0 & 1.0 \end{bmatrix} & \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} & \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \\ \begin{bmatrix} -4.0 & -1.0 & -0.1 \\ 1.0 & -2.5 & -2.0 \\ 3.0 & 2.0 & 1.0 \end{bmatrix} & \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} & \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix} \\ \begin{bmatrix} -4.0 & -1.0 & -0.1 \\ 0 & -2.75 & -2.025 \\ 0 & 1.25 & 0.925 \end{bmatrix} & \begin{bmatrix} 1 & 0 & 0 \\ -0.25 & 1 & 0 \\ -0.75 & 0 & 1 \end{bmatrix} & \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix} \\ \begin{bmatrix} -4.0 & -1.0 & -0.1 \\ 0 & -2.75 & -2.025 \\ 0 & 0.0 & 0.0045454 \end{bmatrix} & \begin{bmatrix} 1 & 0 & 0 \\ -0.25 & 1 & 0 \\ -0.75 & -0.45454 & 1 \end{bmatrix} & \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix} \\ U & L & P \end{array}$$

(b) The elements of the vector $\mathbf{b}$ are stated with five significant digits. If we solve the system $A\mathbf{x} = \mathbf{b}$, how many significant digits can we expect to find for the elements of $\mathbf{x}$? The condition number of A is 2103.2.

The condition number is of order 10^3 , so we lose at most 3 significant digits, thus we expect to find two significant digits for $\mathbf{x}$.

3. [15 marks] Consider the following function:

$$E(x) = \int_{z=0}^x \frac{1}{\sqrt{2 - \sin^2(z)}} dz$$

Your task is to create a table of approximate function values for $x \in [0, \pi]$. An outline of the algorithm is as follows:

1. Let x_j ($j = 0, \dots, n$) be regularly spaced grid points so that $x_0 = 0$ and $x_n = \pi$.
2. For each grid point x_j ($j = 1, \dots, n$), approximate $E(x_j)$ using the composite midpoint rule on j sub domains and add the approximation E_j to the table.

(a) Write a pseudo-code for this algorithm.

Input: The number of sub intervals n .

$T \leftarrow (n+1) \times 2$ array of floats

$h \leftarrow \pi/n$

$x_j \leftarrow jh$ for $j = 0, \dots, n$

Set the first row of T to $(x_0, 0)$

$I \leftarrow 0$

For j from 1 to n :

$m \leftarrow (x_{j-1} + x_j)/2$

$I \leftarrow I + hf(m)$

1 FLOP
($n+1$) FLOPs

2 FLOPs
 $\alpha + 2$ FLOPs

Set the j -th row of T to (x_j, I)

Output: T

For full marks, make sure that each function value is computed only once.

- (b) Compute the number of FLOPs necessary to complete the algorithm as a function of n .

$$\text{\#FLOPs} = n + 2 + \sum_{j=1}^n (2 + \alpha + 2) = 2 + (\alpha + 5)n = O(n)$$

- (c) Suppose we want to make a more accurate table by using the composite Simpson's rule instead of the composite midpoint rule while using the same number of sub intervals. Answer YES or NO to each of the following questions:

- A. Will this change the FLOP count of your algorithm? YES, it will approximately double.
 B. Will this change the order of complexity of your algorithm? NO, the order of complexity will still be $O(n)$.

4. [15 marks] The function

$$f(x) = \frac{1}{1+x}$$

attains the following values

k	1	2	3
x_k	0	1/2	1
y_k	1	2/3	1/2

We interpolate f by a second order polynomial on these nodes.

- (a) Write down the interpolation conditions.

Let the interpolant be $P(x) = a_0 + a_1x + a_2x^2$, then

$$P(x_1) = a_0 = y_1 = 1$$

$$P(x_2) = a_0 + \frac{1}{2}a_1 + \frac{1}{4}a_2 = y_2 = \frac{2}{3}$$

$$P(x_3) = a_0 + a_1 + a_2 = y_3 = \frac{1}{2}$$

- (b) Solve the interpolation conditions and write down the interpolating polynomial.

We get $a_0 = 1$, $a_1 = -5/6$ and $a_2 = 1/3$ and

$$P(x) = 1 - \frac{5}{6}x + \frac{1}{3}x^2$$

- (c) Give an upper bound for the interpolation error. Make sure your upper bound is less than 1.

$$|P(x) - f(x)| = \left| \frac{1}{3!} \frac{-6}{(1+\xi)^4} x \left(x - \frac{1}{2}\right)(x-1) \right| = \left| \frac{1}{(1+\xi)^4} x \left(x - \frac{1}{2}\right)(x-1) \right| \leq$$

$$\max_{\xi \in [0,1]} \frac{1}{(1+\xi)^4} \times \max_{x \in [0,1]} \left| x \left(x - \frac{1}{2}\right)(x-1) \right| = \frac{\sqrt{3}}{36} \approx 0.0482$$

where I have used that the function of ξ is monotonically decreasing. For the second part, just take the derivative with respect to x and equate it to zero. You will find that the maxima are located at $1/2 \pm \sqrt{3}/6$.

5. [15 marks] The following interpolation data are given:

k	0	1	2
x_k	-1.00	0.000	1.00
y_k	0.53	3.80	5.18

- (a) Compute the free, cubic, spline interpolant for these data.

Let the interpolant be given by

$$S(x) = \begin{cases} S_0(x) = a_0 + b_0(x+1) + c_0(x+1)^2 + d_0(x+1)^3 & \text{for } -1 \leq x < 0 \\ S_1(x) = a_1 + b_1x + c_1x^2 + d_1x^3 & \text{for } 0 \leq x < 1 \end{cases}$$

The conditions are

$$\begin{aligned} S_0(-1) &= a_0 = 0.53 & S'_0(0) &= b_0 + 2c_0 + 3d_0 = b_1 = S'_1(0) \\ S_0(0) &= a_0 + b_0 + c_0 + d_0 = 3.80 & S''_0(0) &= 2c_0 + 6d_0 = 2c_1 \\ S_1(0) &= a_1 = 3.80 & S''_0(-1) &= 2c_0 = 0 \\ S_1(1) &= a_1 + b_1 + c_1 + d_1 = 5.18 & S''_1(1) &= 2c_1 + 6d_1 = 0 \end{aligned}$$

from which we have immediately $a_0 = 0.53$, $a_1 = 3.80$ and $c_0 = 0$. The remaining equations are

$$\begin{bmatrix} 1 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 1 & 1 \\ 1 & 3 & -1 & 0 & 0 \\ 0 & 6 & 0 & -2 & 0 \\ 0 & 0 & 0 & 2 & 6 \end{bmatrix} \begin{bmatrix} b_0 \\ d_0 \\ b_1 \\ c_1 \\ d_1 \end{bmatrix} = \begin{bmatrix} 3.27 \\ 1.38 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

Use Gauss elimination or any other method you prefer to find $b_0 = 3.7425$, $d_0 = -0.4725$, $b_1 = 2.325$, $c_1 = -1.4175$ and $d_1 = 0.4725$.

- (b) Suppose we are given three more data points and now want to compute the free, cubic spline interpolant on five intervals. How many interpolation and smoothness conditions do we have to solve?

Each cubic spline component has 4 parameters, so there are 20 in total and 20 conditions to match.

6. [15 marks] Suppose a device records the following (x, y) data pairs:

$$(0.25, 0.6137), (0.50, 1.307), (0.75, 1.712), (1.0, 2.000), \text{ and } (1.25, 2.223)$$

Assume that $y = f(x)$ for some unknown function f .

- (a) Approximate $f'(0.25)$ using forward differences.

$$f'(0.25) \approx \frac{f(0.50) - f(0.25)}{0.25} = 2.7732 \dots$$

- (b) Approximate $f''(1.0)$ using centered differences.

$$f''(1.0) \approx \frac{f(0.75) - 2f(1.0) + f(1.25)}{0.25^2} = -1.0400 \dots$$

- (c) If we use all data to obtain an estimate for $f'(0.75)$, then what is the highest possible order of the error in terms of the distance h between grid points?

Order $O(h^4)$. We can tell from the error of polynomial interpolation. Its polynomial part is $(x - x_{-2})(x - x_{-1})(x - x_0)(x - x_1)(x - x_2)$ (if we label the x -values so that $x_0 = 0.75$. Taking the derivative and evaluating at $x = x_0$, we find the stated order.

7. [10 marks] Approximate

$$\int_0^1 \ln(1+t^2) dt$$

using the composite Simpson's rule with two subintervals of equal size.

The first subinterval is $[0, 1/2]$ with mid point $1/4$ and the second subinterval is $[1/2, 1]$ with mid point $3/4$, therefore

$$I = \frac{1}{12} \left(\ln(1+0) + 4 \ln\left(1 + \frac{1}{16}\right) + \ln\left(1 + \frac{1}{4}\right) \right) + \frac{1}{12} \left(\ln\left(1 + \frac{1}{4}\right) + 4 \ln\left(1 + \frac{9}{16}\right) + \ln(1+1) \right) = 0.2639 \dots$$

8. [5 marks] For each of the following statements, indicate if they are true or false:

(a) If a quantity y approximately depends on a quantity x through an exponential law (meaning that $y \approx a \exp(bx)$ for some constants a and b) we should plot data of y versus x on a semilogarithmic scale (lin-log) to see the data fall approximately on a straight line.

True

(b) If we use Newton iteration to find the root of a function $f(x)$, the quantity $|1/f'(\bar{x})|$ can be considered a condition number for this iterative method.

True

(c) Numerical integration (quadrature) is well-behaved (stable and accurate) when taking the limit of the grid spacing to very small values (e.g. 10^{-14} when using the Python "float" data type).

True

(d) A recursive implementation of algorithms (in the form of a function that calls itself) is always more efficient than an iterative one (in the form of a function with explicit loops).

False

(e) For each numerical algorithm we develop, we must specify

1. under what conditions it will work,
2. how accurate the result will be and
3. how many operations it will take to complete (at least the order of complexity).

True

9. [10 marks (bonus)] Suppose we take measurements of a quantity that is approximately periodic in time - like the outside temperature from day to day or the water height in lake Ontario from high tide to high tide. We want to compute a least-squares approximant to the data that is periodic. Assume that the period is 2π , then our approximant is

$$G(t) = a \sin(t) + b \cos(t)$$

Let t_j ($j = 0, \dots, n$) be equally spaced time intervals do that $t_0 = -\pi$ and $t_n = \pi$. Derive an expression for the coefficients a and b in terms of the input data (x_j, y_j) .

The error is

$$E = \sum_{j=0}^n (y_j - a \sin(x_j) - b \cos(x_j))^2$$

To find the minimum, we equate the derivatives to zero:

$$\frac{\partial E}{\partial a} = -2 \sum_{j=0}^n (y_j - a \sin(x_j) - b \cos(x_j)) \sin(x_j) = 0$$

$$\frac{\partial E}{\partial b} = -2 \sum_{j=0}^n (y_j - a \sin(x_j) - b \cos(x_j)) \cos(x_j) = 0$$

which we can expand to

$$\sum_{j=0}^n (y_j \sin(x_j) - a \sin^2(x_j) - b \sin(x_j) \cos(x_j)) = 0$$
$$\sum_{j=0}^n (y_j \cos(x_j) - a \cos(x_j) \sin(x_j) - b \cos^2(x_j)) = 0$$

now the sums over $\sin(x_j) \cos(x_j)$ equal zero because the contribution at grid point x_j is cancelled by the contribution at grid point $-x_j$. Finally, we have

$$a = \frac{\sum_{j=0}^n y_j \sin(x_j)}{\sum_{j=0}^n \sin^2(x_j)} \quad b = \frac{\sum_{j=0}^n y_j \cos(x_j)}{\sum_{j=0}^n \cos^2(x_j)}$$